



Slovenian law AI assistant

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Abstract

Large language models (LLMs) are powerful but often lack the domain-specific knowledge required for accurate legal advising, particularly in localized contexts such as Slovenian law. In this work, we outline the design of a specialized conversational AI assistant aimed at addressing these limitations. The proposed approach combines the COESLAW 1.0 legal corpus with techniques such as Retrieval-Augmented Generation (RAG) and parameter-efficient fine-tuning (PEFT) to enable context-aware and reliable responses grounded in authoritative sources.

Keywords

AI assistant, Slovenian legislation, Retrieval-Augmented Generation

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Introduction

Although modern large language models are capable of generating fluent and persuasive responses, they are not inherently equipped to handle highly specialized, institution-specific tasks. Their training is based on broad and often globally sourced datasets, which limits their ability to provide precise, localized, and up-to-date information. As a result, they may produce incomplete, inaccurate, or even hallucinated answers when faced with context-dependent queries.

This project addresses these limitations by developing a specialized conversational AI assistant focused on legal advising within Slovenian law. Rather than relying solely on general-purpose knowledge, the system aims to incorporate domain-specific expertise through structured knowledge bases and targeted adaptation techniques.

Related work

In recent years, a growing number of AI-based legal assistants have been developed, particularly in English-speaking countries. One of the most well-known examples is DoNotPay, an AI-powered legal service that automates tasks such as contesting parking tickets and generating legal documents [1]. Another example is PainWorth, a Canadian platform that uses AI to automate personal injury claims [2]. More advanced systems such as Harvey AI employ large language models to assist legal professionals with research, document drafting, and compliance review [3]. These systems illustrate a broad spectrum of AI applications in law, ranging from consumer-

facing tools to professional-grade platforms.

In Slovenia, such systems are less common. One notable example is Pravko [4], an AI-based assistant designed to help users navigate legal and administrative procedures. Pravko understands natural language queries and retrieves answers from official sources such as Slovenian legislation and e-government portals. However, existing systems remain limited, highlighting the need for more advanced, data-driven approaches based on large-scale legal datasets and modern language models, which is the focus of our work.

Methods

By using approaches such as Retrieval-Augmented Generation (RAG) and parameter-efficient fine-tuning (PEFT), the assistant will be designed to interpret user intent, retrieve relevant legal information, and generate accurate, context-aware responses. The ultimate goal is to create a reliable AI system that can function as a domain-specific expert, minimizing hallucinations while providing trustworthy and precise guidance.

COESLAW 1.0: Database of Slovene legal texts

For the development of our assistant, we use the COESLAW 1.0 corpus [5], a large-scale collection of Slovenian legal texts. The dataset was published in 2024 as part of the LLM4DH project [6] and is specifically designed to support research in legal natural language processing and domain adaptation of language models.

The corpus contains 547,799 documents with a total of approximately 772 million words. It aggregates documents from

multiple authoritative public sources, including the Legal Information System of the Republic of Slovenia (PISRS), court decisions (SodnaPraksa), the Constitutional Court (USRS), and the Official Gazette (Uradni List).

The dataset covers a wide range of legal domains, including enacted and repealed laws, legislative proposals, lower court decisions, regulatory and announcement. Documents are stored in a structured JSONL format and include full cleaned text along with source-specific metadata such as document identifiers, titles, legal basis references, document types and date of decision.

Evaluation

The performance of the system will be evaluated using both qualitative and comparative approaches. We will compare the generated answers with existing legal question-answering platforms, such as publicly available answers from Moj Mali Pravniki [7] and the Pravko [4] system.

Evaluation will focus on:

- **Accuracy:** correctness of the legal information
- **Source support:** whether the answer is supported by retrieved documents
- **Clarity:** readability and usefulness of the response

Results

Discussion

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